Problem Set 6: Sponsored Search Auctions

3. The Balanced-Bidding Agent

3.1 Team name

Our team name is **cwjc**.

3.2 Strategy implementation

For each slot, the agent removes its own previous bid, sorts the remaining bids implicitly through the simulator's slot-range helper, and uses the minimum bid for that slot as the predicted GSP price. It estimates the slot's value as previous-round clicks times (value minus predicted price), then targets the slot with greatest estimated utility. If the target is the top slot, or its predicted price is at least the agent's value, the bid is its value. For any lower target slot, it solves the assignment's indifference equation against the immediately higher slot and rounds down to an integer cent, bounded between zero and value. This uses previous-round position clicks as the position-effect estimate and respects the reserve through the bid-range helper.

3.3(a) All truthful versus all balanced

We ran five agents for 48 periods, with the default $5,000 budget, seed 2, 200 value draws, and one sampled value permutation per draw. Utilities below are average daily dollars per agent.

Population	Mean utility / agent	Range among seats	Mean daily revenue
Five Truthful	$337.41	$328.17-$355.38	$6,139.20
Five Cwjcbb	$641.59	$607.02-$675.00	$4,616.72

In this experiment balanced bidders earn substantially more utility and generate less revenue. Truthful bids do not account for the fact that GSP charges the next bidder's bid: by bidding value, an agent can pay more than necessary for a position. Balanced bidding uses observed competition to trade position against price. These figures are estimates for this seed and permutation sampling, not exact expectations.

3.3(b) Mixed populations

For each mixed run we used five agents, seed 2, 200 draws, and 10 sampled value permutations per draw. The output reports average daily utility for each seat.

Population	Agent type	Mean utility / agent
Four Truthful, one Cwjcbb	Truthful (mean over four)	$383.76
Four Truthful, one Cwjcbb	Cwjcbb	$567.85
Four Cwjcbb, one Truthful	Cwjcbb (mean over four)	$644.65
Four Cwjcbb, one Truthful	Truthful	$649.22

The balanced bidder has a sizable advantage when surrounded by truthful bidders. Against four balanced bidders, the single truthful bidder's utility is close to (and in this sample slightly above) the balanced-agent mean. This is consistent with balanced bidding exploiting truthful overbidding while yielding little systematic advantage over other balanced bidders; the modest difference in the second mixed run should not be read as a robust strategy advantage.

4. Revenue: GSP versus VCG

4.1 VCG payment rule

After filtering out bids below reserve and sorting eligible bids, the code computes total payments from the lowest allocated slot upward. The lowest allocated bidder pays its clicks times the larger of reserve and the highest unallocated bid. Each higher bidder's payment adds the externality on the next slot, (clicks in this slot minus clicks in the next slot) times the next bidder's bid, to the lower slot's total payment. Dividing each total by its slot clicks returns the per-click prices expected by the simulator. The payment implementation passes the supplied VCG checks.

4.2(b-c) Reserve-price revenue

We ran the specified reserve grid from 0 through 170 cents in 10-cent increments, with five agents, seed 2, 200 draws, one sampled permutation per draw, and the default budget. Revenue is average daily dollars.

Reserve (cents)	GSP + balanced	VCG + truthful	Reserve (cents)	GSP + balanced	VCG + truthful
0	$4,616.72	$4,345.24	90	$5,201.39	$5,109.71
10	$4,616.72	$4,345.24	100	$4,921.53	$4,857.88
20	$4,616.72	$4,345.24	110	$4,920.91	$4,880.81
30	$4,617.32	$4,358.02	120	$4,570.69	$4,561.00
40	$4,751.97	$4,545.75	130	$4,254.49	$4,243.38
50	$4,687.79	$4,465.68	140	$3,567.80	$3,563.65
60	$4,948.72	$4,746.63	150	$2,809.52	$2,808.90
70	$5,205.38	$5,012.76	160	$1,669.61	$1,669.99
80	$5,175.56	$5,044.00	170	$703.83	$703.88

On this grid the revenue-maximizing reserve is 70 cents for GSP with balanced bidders ($5,205.38) and 90 cents for VCG with truthful bidders ($5,109.71). Revenue initially increases as the reserve screens out low bids and raises the payment floor. At high reserves, fewer bids qualify, reducing filled positions and total revenue. Small non-monotonic movements around the maxima reflect sampling noise and changes in which bidders qualify.

4.2(d) Switching from GSP to VCG at period 24

With balanced agents, reserve zero, seed 2, and 200 draws, revenue in periods 0-23 averaged $2,394.51 per day-half under both the all-GSP and switch runs. In periods 24-47, the all-GSP comparison averaged $2,222.21, while the run that switched to VCG averaged $1,819.06. Thus revenue fell after the switch, by about 18% compared with the same-seed all-GSP second half.

This is economically plausible because these agents choose bids to balance GSP prices against positions. VCG charges each winner for the externality it imposes on other bidders, which differs from the next-bid GSP price and is often lower for higher positions when click-through rates fall down the page. Bidders continue using their GSP-oriented adjustment after the mechanism changes, so bids and payments are no longer adapted to the payment rule. The observed revenue change is therefore specific to these agents and this experiment, not a universal ranking of the mechanisms.

4.2(e) Takeaways

These experiments show that bidder strategy and auction design jointly shape utility and revenue. In GSP, adapting bids to the observed price-position tradeoff can substantially improve bidder utility relative to truthful bidding, while competition from truthful bidders creates an opening for a balanced bidder. Reserves can increase revenue by setting a floor but can also exclude too many bidders; the best tested reserve differs across the two payment rules and bidder populations. Changing from GSP to VCG without changing bidder behavior lowers revenue in this simulation, illustrating that mechanism performance depends on how agents respond to the rules. Because each reserve estimate used one sampled permutation per value draw, the reported optima should be treated as the best grid points in this seeded experiment rather than precise universal optima.

5. Budget-Constrained Competition Agent

Design. Cwjcbudget is self-contained in its submitted Python file. It starts at two-thirds of its value, then observes the previous period's other bids and slot clicks, selects the highest-surplus target slot, and computes the corresponding balanced bid. It compares its observed cumulative spending with a linear daily spending plan. If spending is ahead of plan, it scales down a value-based bid cap; if spending is on or behind plan, it can use its full balanced bid up to value. It returns zero if its remaining budget is exhausted. This simple pacing rule aims to preserve budget for later periods while retaining the position-price logic of balanced bidding. The simulator itself forces zero bids after exhaustion and permits the stated final-period overspend.

Expected behavior. The agent should spend more cautiously when early payments put it ahead of pace, but it does not forecast future opponents or solve a dynamic optimization problem. It may therefore leave some budget unused or spend too quickly when competition changes. The assignment emphasizes a sensible, correctly running budget-aware agent rather than tournament placement.